

42. Scripts: async aur defer

Problem

Browser jab HTML me `<script>` dekhta hai to DOM banana **rok** deta hai aur script download + execute karta hai. Isse do dikkatein:

1. Script apne **neech ke DOM elements nahi dekh sakti**.
2. Upar bhaari script ho to **page block** ho jaata hai (user ko content nahi dikhta).

Purana workaround: script `<body>` ke bilkul neech rakho. Par ye perfect nahi (browser script ko HTML poora download hone ke baad hi dekhta hai, lambe HTML me deri).

Solution: `defer` aur `async` attributes.

`defer`

- Script background me download hoti hai, HTML/DOM banana **nahi rukta**.
- DOM **poora ban jaane par** chalti hai (par `DOMContentLoaded` se pehle).
- **Order safe rehta hai** (document me jaisa likha, waisa hi execute), chahe chhoti script pehle download ho jaye.
- `DOMContentLoaded` deferred scripts ka wait karta hai.
- Sirf **external** scripts ke liye (`src` ke bina ignore).

html

```
<script defer src="long.js"></script>
<script defer src="small.js"></script>    <!-- long.js ke baad hi chalegi -->
```

`async`

- Script **poori tarah independent**: koi kisi ka wait nahi karta.
- DOM/dusri scripts async ka wait nahi karti, aur async unka nahi.
- `DOMContentLoaded` async ke pehle ya baad kabhi bhi aa sakta hai (guarantee nahi).
- **Load-first order**: jo pehle download hui wo pehle chalegi.
- Kaam ki jagah: counters, ads, analytics jaise independent third-party scripts.

html

```
<script async src="https://google-analytics.com/analytics.js"></script>
```

- Ye bhi sirf external scripts ke liye.

Dynamic scripts

js

```
let script = document.createElement('script');
script.src = "/long.js";
document.body.append(script); // append hote hi load shuru
```

- Ye **default me** `async` jaisi behave karti hain (load-first).
- Order chahiye to `script.async = false` set karo (tab `defer` jaisa document order):

js

```
function loadScript(src) {
  let script = document.createElement('script');
  script.src = src;
  script.async = false;
  document.body.append(script);
}
```

Farak (summary table)

Order	<code>DOMContentLoaded</code>
<code>async</code> Load-first, document order matter nahi	Irrelevant, pehle ya baad kabhi
<code>defer</code> Document order	Document parse hone ke baad, <code>DOMContentLoaded</code> se just pehle

Kab kya: `defer` = jab poora DOM chahiye ya scripts ka order matter kare (library + uspar depend script). `async` = independent scripts.

Dhyan rakho

`defer/async` se user page **script load hone se pehle** dekh leta hai. Isliye "loading" indicator dikhao aur jo buttons abhi kaam ke nahi unhe disable rakho.